

UN DECADE ON ECOSYSTEM RESTORATION

REVIEW ARTICLE

Restoring tropical forest–grassland mosaics invaded by woody exotics

Manaswi Raghurama^{1,2} , Mahesh Sankaran^{1,3}

Although important for biological conservation and provisioning of several ecosystem services, tropical forest–grassland mosaics face several threats, one of which is woody invasion in grasslands of these mosaics—an issue proving to be challenging to solve. We present a framework to guide grassland restoration in these mosaics that takes into account their unique ecology, outlines a general chronological sequence of restoration activities, and helps identify research priorities to guide restoration efforts. Specifically, we outline approaches to (1) increase the cost-effectiveness of restoration programs (e.g. selecting restoration targets and priority sites); (2) restore edaphic processes impacted by woody invasion; (3) maximize the establishment of native grass and forb species, and (4) maintain restored communities in the long-term by minimizing re-invasions. We apply this framework to montane southern Indian forest–grassland mosaics (locally called shola–grassland mosaics) invaded by nitrogen-fixing woody exotics to identify research priorities that can act as “game-changers” in the restoration of these ecosystems. These include (1) identifying invaded areas where passive restoration can be effective, (2) understanding the role of endogenous disturbances, particularly fire, in the spread and management of invasive species and the prevention of re-invasion, and (3) assessing the role of large-scale seed farming and emerging seed technologies for the use of seeds to effectively re-introduce native grassland species. Restoration frameworks that take into account unique ecologies are needed to make the most of the UN Decade on Ecosystem Restoration, and we present such a framework for forest–grassland mosaics—an important ecosystem that is generally neglected.

Key words: forest–grassland mosaics, Nilgiris, plant invasion, shola–grasslands, UN decade on ecosystem restoration, woody exotics

Conceptual Implications

- Management efforts in tropical forest–grassland mosaics should prioritize conservation of uninvaded areas and target the invasion front before tackling heavily invaded areas. Current restoration efforts of invaded grasslands in these mosaics, which mostly amount to clearing invasive biomass, are unlikely to succeed, with re-invasions common after such efforts.
- Cost-effective restoration will require managers to appreciate the unique ecology of these mosaics, account for impacts of woody invasives on edaphic properties and soil seed banks, and gain a better understanding of native and invasive species biology and the role of endogenous disturbances to maximize native species re-establishment and minimize re-invasion in restored communities. Involvement of local communities can provide economic benefits and improve livelihoods, while also increasing the cost-effectiveness of restoration efforts.

grassland matrix (Bond & Parr 2010; Parr et al. 2014). Tropical and sub-tropical FG mosaics around the globe (many in global biodiversity hotspots; León Cordero 2016) harbor unique and high biodiversity (e.g. Overbeck et al. 2007; Bond et al. 2008), and provide several ecosystem services (ES) including sequestering carbon (C) in stable forms in soil, regulating water flow, providing forage for livestock and habitat for pollinators (Zhao et al. 2020). Although historically believed to be anthropogenic in origin, a growing body of evidence suggests that many of these mosaics are natural and ancient, and maintained by different processes (e.g. fire, herbivory, edaphic properties, and frost, among others) that promote one ecosystem state over the other in different parts of the landscape (Warman & Moles 2009; Joshi et al. 2020).

FG mosaics are threatened by several anthropogenic drivers including land conversion for crops, forestry, urbanization, and invasion by exotic woody species (Bond & Parr 2010; Joshi et al. 2020). Climate change poses an additional threat to these

Author contributions: MR, MS jointly conceived the idea for this work, wrote and revised the manuscript, and gave final approval.

¹Ecology & Evolution Group, National Centre for Biological Sciences (NCBS), Tata Institute of Fundamental Research, Bengaluru, Karnataka 560065, India

²Address correspondence to M. Raghurama, email manaswi1993@gmail.com

³School of Biology, University of Leeds, Leeds, U.K.

Introduction

Forest–grassland (FG) mosaics are natural complexes characterized by distinct patches of C_3 trees embedded within a C_4

© 2021 Society for Ecological Restoration.
doi: 10.1111/rec.13491

mosaics, with the potential to drive future biome switches from grasslands to forests (Joshi et al. 2020). Grasslands of these mosaics have typically been more negatively impacted by human activities than their forest counterparts (e.g. Overbeck et al. 2007; Arasumani et al. 2019), largely due to a historical lack of understanding and acknowledgment of the importance of these old-growth grasslands (Veldman et al. 2015a; Bond 2019; Silveira et al. 2020). Grasslands can contribute 33–50% of the diversity of these mosaics (Bond & Parr 2010), and in some cases can be more diverse than their forest counterparts (Nangendo et al. 2002). A failure to recognize the unique ecology of these grasslands, and integrate these into international restoration programs such as the Bonn Challenge (Veldman et al. 2017; Buisson et al. 2019; Temperton et al. 2019), has resulted in ill-conceived management policies, mostly amounting to large-scale afforestation of grasslands by woody species, including exotics (Joshi et al. 2018; Bond 2019). Invasion by exotic woody species is a particularly widespread threat to FG mosaics globally—they decrease native grassland diversity, and alter nutrient cycling processes and the provisioning of several critical ES (Veldman et al. 2015b).

In this paper, we present a framework to guide restoration programs and prioritize research to better inform restoration efforts in invaded grasslands of tropical FG mosaics. In the context of the UN Decade on Ecosystem Restoration, frameworks that take into account each ecosystem's unique ecology are needed to guide their restoration (Dudley et al. 2020). Our framework complements, and builds on, recently published frameworks for the restoration of tropical grassy biomes (TGBs; Buisson et al. 2019; Buisson et al. 2020). We finally

apply this framework to the case of restoring grasslands invaded by woody Nitrogen (N)-fixing exotics in the montane *shola*-grassland mosaics of southern India.

A Framework for Restoration of Grasslands Invaded by Woody Exotics in Tropical FG Mosaics

Restoration of grasslands invaded by woody species has been previously conceptualized as a “wicked problem”—complex and expensive to solve, and needing adaptive management (Le Maitre et al. 2011; Woodford et al. 2016). Large-scale eradication of woody species can be extremely expensive and difficult (van Wilgen & Richardson 2014), and containing invasives from expanding further into uninvaded areas should, therefore, generally be the first priority in these grasslands (Le Maitre et al. 2011). We, however, focus here on enabling the cost-effective eradication of woody invasives and the restoration of invaded grasslands, while recognizing that preventing further invasion should be the first line of defense (Fig. 1).

Broadly, our framework emphasizes the need for (1) appropriate site-selection and restoration approaches to increase cost-effectiveness, (2) quantifying the impacts of invasive species on edaphic processes so that they can be effectively restored, (3) understanding the biology of native grassland species across multiple life stages, particularly the seed and seedling stages, to maximize their establishment, and (4) understanding the role of native plant functional traits, diversity and endogenous disturbances in preventing reinvasions and ensuring the long-term persistence of restored communities. Setting clear restoration targets, identifying priority areas and areas where passive restoration can be effective, and identifying cost-effective methods to

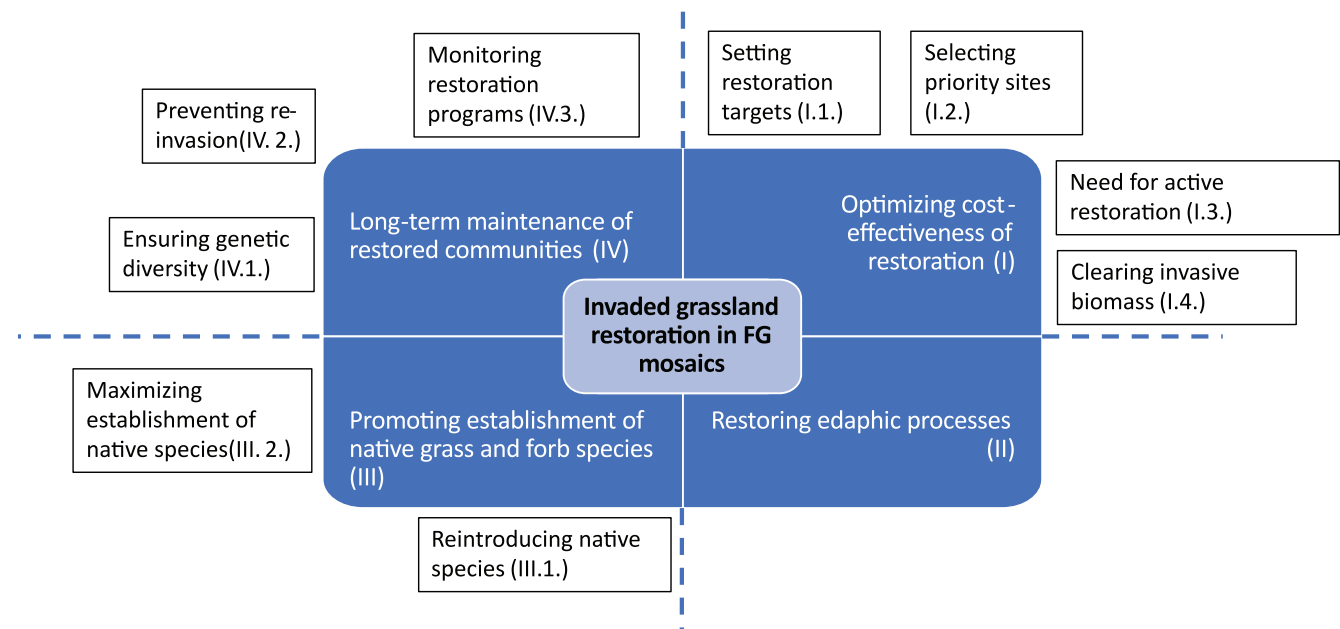


Figure 1. Framework to restore grasslands invaded by woody exotics in tropical FG mosaics. Generally, planning and execution will temporally follow a clockwise sequence, starting from the top-right quadrant. We grouped various aspects (white boxes) of the framework under four stages (blue boxes). Numbers in brackets denote the stages and substages where each of these aspects are discussed below.

remove invasive biomass, should be the first step in the restoration process; one that can significantly improve the cost-effectiveness of restoration programs in FG mosaics (stage I). Because woody invasion is known to significantly impact edaphic states and processes (Jo et al. 2017), restoring these is likely to be a prerequisite to encourage the establishment of native species, prevent secondary invasion, and restore soil- and water-related ecosystem services following invasive biomass removal (stage II). Reintroduction of native grassland species should not only be aimed at improving native diversity, but also at restoring ecosystem functioning, while activities to maximize the establishment of reintroduced species is needed to scale-up restoration in these ecosystems (stage III). Finally, ensuring genetic diversity of restored communities, preventing re-invasion, and monitoring restored communities to facilitate adaptive management is required to ensure the long-term maintenance of these communities (stage IV).

Optimizing Cost-Effectiveness of Restoration (Stage I)

I.1. Setting Restoration Targets

Restoration targets, that is the desired state post-restoration, can differ based on ecological, economic, and political contexts, and can range from a focus on restoring biodiversity, or on improving ecosystem functioning and provisioning of ES (Perring et al. 2015). Restoration targets or “reference states” can be uninvaded native grasslands (Durbecq et al. 2020), historical grassland states based on written records and/or paleoecology (Manzano et al. 2020), “novel ecosystems” based on what is realistically achievable (Hobbs et al. 2009), or resilient to predicted climate change scenarios (Gann et al. 2019), or a combination of these (Prober et al. 2015). Further, to ensure the long-term sustainability of restored communities, they also need to be resilient to endogenous disturbances (e.g. fire, herbivory; Buisson et al. 2020).

I.2. Selecting Priority Areas for Restoration

Priority sites for restoration must be based on several factors that determine the probability of restoration success (e.g. invasion intensity, habitat connectivity; Wilson et al. 2011; Tambosi et al. 2014; Rappaport et al. 2015), and its cost-effectiveness (e.g. site accessibility). Further, understanding the drivers of FG mosaics and targeting grassland restoration in sites where environmental drivers do not facilitate woody encroachment will help in the long-term maintenance of the restored communities, and recreate multifunctional FG mosaics (Buisson et al. 2020). In areas where environmental conditions promote forests, it is better to restore native forests, if needed (however, more details on this are outside the scope of this study). For example, frost, which occurs more often in depressions and valley bottoms (Bader & Ruijten 2008), decreases the survival of native trees in montane grasslands (Wakeling et al. 2012; Joshi et al. 2020). Therefore, priority sites for grassland restoration from invasion in such mosaics should be in topographic depressions and valley bottoms. Once prioritization criteria are

selected, spatial information layers corresponding to these criteria can be used in multi-criteria spatial prioritization tools (e.g. ARIES [Artificial Intelligence for ES]; Villa et al. 2014) to help identify areas in the landscape where restoration efforts would deliver greatest benefits given limited resources (Strassburg et al. 2019).

I.3. Evaluating Need for Active Restoration

While active restoration may be necessary in sites where invasion intensity is high (Buisson et al. 2019; Holmes et al. 2020; Fig. 2), identifying sites where passive restoration can be effective, such as sites with existing native vegetation and where underground storage organs (USOs) and seed banks of native vegetation are not significantly degraded (Fensham et al. 2016; Prach et al. 2019a), can significantly improve the cost-effectiveness of restoration programs.

I.4. Clearing Invasive Biomass

Woody invasives are usually cleared manually, but this activity is resource-intensive and can impact soil stability and promote secondary invasions (Pearson et al. 2016). Controlled fires can be used to clear woody biomass in many cases (Grace et al. 2001), as well as to deplete soil seed banks of invasive plants (DiTomaso et al. 2006), and to volatilize excess nutrients (Blair 1997). On the other hand, fires can also encourage secondary invasion in some cases, particularly if fire breaks invasive seed dormancy or if invasives regenerate post-fire, and employing other techniques such as manual weeding, mowing and herbicide application may be needed to prevent re-invasion in such cases (Nsikani et al. 2019; Sriramamurthy et al. 2020). Finally, clearing invasives can be an employment-generating activity, and cleared biomass can be used as a resource by local communities or businesses to increase cost-effectiveness further (e.g. *Acacia* for fuelwood, *Lantana* for furniture, Kull et al. 2011; Kannan et al. 2016). However, clearing of woody invasives is an intensive exercise that can cause significant disturbance, and so care should be taken, and appropriate protocols developed, to minimize soil disturbance and ensure native USOs are not destroyed.

Restoring Edaphic Processes (Stage II)

Woody invasion, particularly by N-fixing species, can fundamentally transform soil characteristics and soil microbial community composition, and as a result change soil microbial activity, and increase nutrient availability and cycling rates (Inderjit & Van Der Putten 2010; Vilà et al. 2011; Jo et al. 2017; Fig. 3), especially in oligotrophic systems (Vitousek & Walker 1989; Sardans et al. 2017). These changes can persist after clearing of invasives (legacy effects; Grove et al. 2015) and create barriers to the restoration of native vegetation (Nsikani et al. 2018). Techniques to reduce soil nutrients post-clearing, such as removal of litter and woody biomass, mowing and exporting biomass, carbon addition, and grazing may be needed (Nsikani et al. 2018) to facilitate re-establishment of

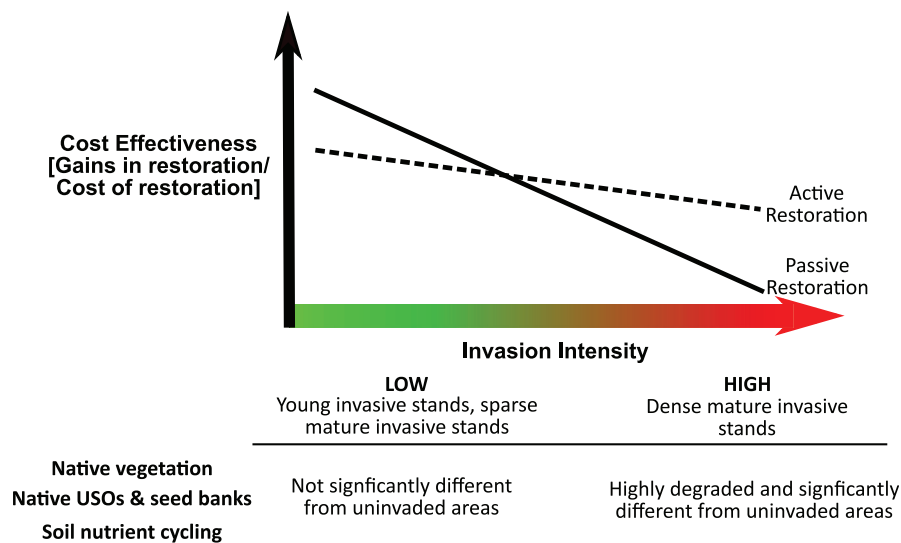


Figure 2. Although cost-effectiveness of restoration can decrease with increasing invasion intensity, the relative cost-effectiveness of active restoration is higher as compared to passive restoration at higher invasion intensities, with passive restoration being almost ineffective in areas with high-invasion intensity. Text below the graph outlines what low- and high-invasion intensities mean, and how certain community and ecosystem characteristics change in response to different invasion intensities.

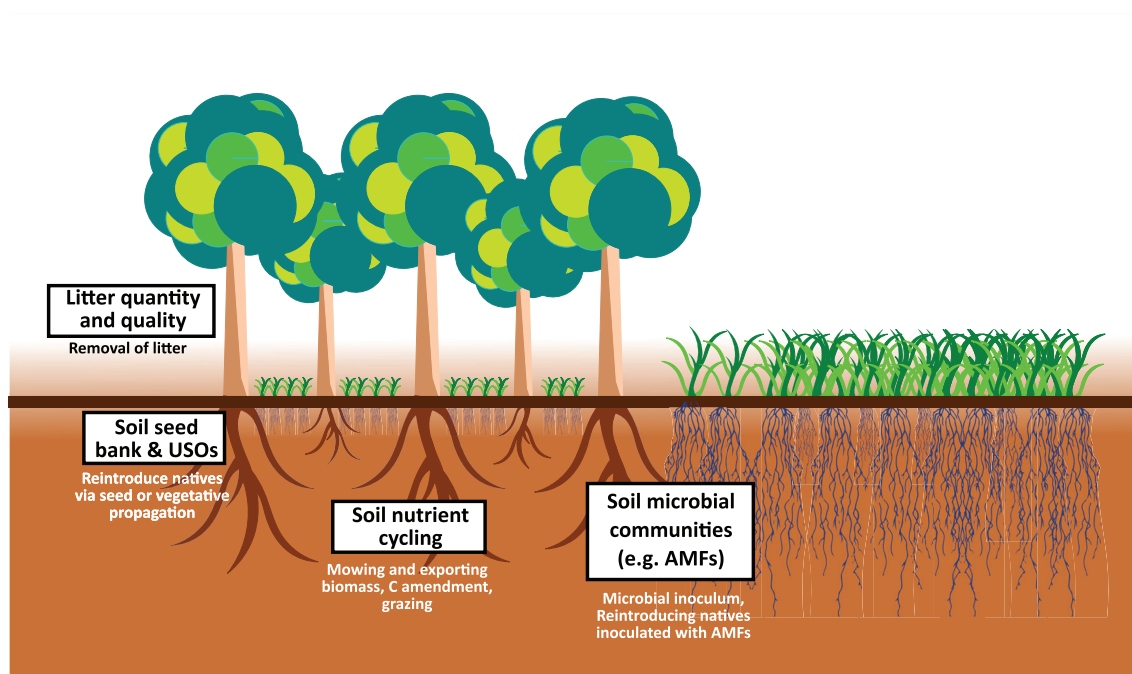


Figure 3. Potential impacts of woody invasion in grasslands of FG mosaics on edaphic properties and suggested activities to reverse these impacts. Text in rectangular boxes are edaphic properties that are directly or indirectly impacted by woody invasion, and text below those boxes (in white color in some cases to increase clarity) are examples of activities to restore these processes. Illustration by Aditi Mishra. AMFs, Arbuscular mycorrhizal fungi.

native species. Planting fast-growing native grasses that sequester nutrients and removing their biomass in a couple of years can help reduce soil nutrients, while employing erosion barriers can promote initial soil stability following clearance (Perry et al. 2010; Kimiti et al. 2017). Subsequent replacement of these fast-growing species with slow-growing perennials adapted to

low nutrient conditions can help maintain low nutrient levels in the longer term (Perry et al. 2010). Further, the efficacy of using microbial inoculum or microbial inhibitors (e.g. nitrification inhibitors) to remediate plant–soil feedbacks should be explored further (Eviner & Hawkes 2008; Perkins & Hatfield 2016). For example, studies have shown that

mycorrhizal fungal and non-mycorrhizal microbial inoculum have positive impacts on the performance of C₄ grasses and non-N-fixing forbs, but not on N-fixing woody species (see Hoeksema et al. 2010 for a meta-analysis).

Promoting Establishment of Native Grass and Forb Species (Stage III)

III.1. Reintroducing Native Species

Since propagation from seeds is difficult for TGB species (Dayrell et al. 2016; Bond 2019; Buisson et al. 2020), grassland restoration might require using vegetative propagation techniques such as transplanting seed-germinated grasses or individual tillers grown in pots (du Toit 2009; de Oliveira et al. 2020), and propagation from bulbs, roots or micropropagation for herbaceous species (Biondo et al. 2007; Hoffman & Smith 2018; more references in Buisson et al. 2020). Although vegetative propagation can be a successful strategy in small-scale restoration efforts (Figueiredo et al. 2018), it still requires the establishment and management of large nurseries, and significant scientific and technical expertise. Studies aimed at improving restoration outcomes from seed, such as identifying dormancy-breaking mechanisms for TGB grasses and herbs (Fig. 4), are potential game-changers that can help significantly increase the cost-effectiveness and ease of doing restoration in these ecosystems.

In cases where recreating the original grassland community is impractical, efforts can focus on restoring grassland functionality by using a subset of native species that are most likely to aid in the restoration of key processes or ecosystem variables of interest (Tsujii et al. 2020). For example, if woody invasion results in less palatable grasses and/or altered fire regimes (Brooks et al. 2004; Holmes et al. 2020), palatable species with flammability similar to the original grassland community (see Wigley et al. 2021 for traits' list) can be targeted for restoration. Functional traits of grasses and forbs can also be used to shortlist

potential native species that could be effective competitors against invasive seedlings (Funk et al. 2008).

III.2. Maximizing Establishment of Reintroduced Native Species

While initially introducing fast-growing native grasses can help prevent soil erosion following clearance and outcompete emerging invasive seedlings (e.g. Coutinho et al. 2019), they can also prevent the subsequent establishment of other slow-growing perennials (Nerlekar & Veldman 2020). Experimental studies can help us understand whether slow-growing perennials should be introduced initially (Fynn et al. 2009). Using nurse plants and restoring soil microbes can also improve the establishment of target native species (Gómez-Aparicio 2009; Hoeksema et al. 2010). In particular, arbuscular mycorrhizal fungi (AMFs), whose diversity is known to decrease due to invasion (Grove et al. 2017), are root symbionts known to improve restoration success in several invaded grasslands (Koziol et al. 2018). Native AMFs can improve the growth of slow-growing late-successional species (Koziol & Bever 2015), although the impact of re-introduction of particular AMFs on the establishment of such species in tropical grasslands remains to be tested (Koziol & Bever 2017).

Long-Term Maintenance of Restored Communities (Stage IV)

IV.1. Ensuring Genetic Diversity of Restored Communities

Using propagules with low genetic variability and/or non-local provenances in restoration sites can reduce establishment success of native species and limit the ability of restored communities to adapt to climate change (Williams & Davis 1996; McKay et al. 2005; Bucharova et al. 2017). Increasing genetic diversity of restored communities by using tillers and seeds collected from spatially separated and local populations, and planting propagules from genetically differentiated populations in close

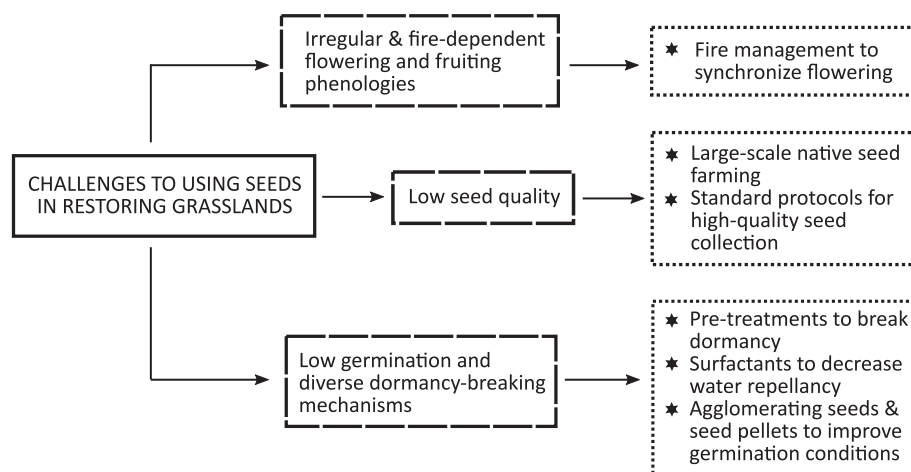


Figure 4. A collection of upcoming research efforts around the globe that can help increase the effectiveness of using seeds in restoration of TGB grass and forb species. Text within dashed boxes are challenges in using seeds in restoring these systems due to the traits of these species. Text within dotted boxes are the research efforts that can help mitigate these challenges. See Dayrell et al. (2016), Madsen et al. (2016), and Wagenius et al. (2020) for more information.

proximity to each other can facilitate sexual reproduction and enhance restoration success (Sommerville et al. 2013; Rogers & McGuire 2015).

IV.2. Preventing Re-Invasion of Restored Communities

Reintroducing native species with traits and phenology similar to those of invasives, that is employing limiting similarity, can increase invasion resistance of restored communities (Pokorny et al. 2005; Funk et al. 2008; Cleland et al. 2013). Reintroducing endogenous disturbances such as fire and herbivory can prevent woody re-invasion in many (*sensu* Midgley et al. 2010), but not in all cases (e.g. Le Maitre et al. 2011), and a site-specific understanding of combinations of different disturbances that prevent re-invasion is needed.

IV.3. Monitoring Restoration Programs

Monitoring the progress and impact of grassland restoration programs is particularly important since re-assembly of native grassland communities can often take centuries (Nerlekar & Veldman 2020). The establishment of comprehensive monitoring programs allows practitioners to periodically evaluate the success of restoration efforts and apply course corrections as needed. Monitoring schedules and indicators need to be established during the initial phases of project planning (Prach et al. 2019b) and should include ecological and biophysical (e.g. biodiversity, soil quality, C-sequestration, hydrological services, among others; Stanturf et al. 2014), as well as socio-economic variables (e.g. livelihood generation and impacts on community welfare; Wortley et al. 2013). Specifically for TGBs, indicators such as recovery after fire and herbivory have also been proposed as measures for monitoring restoration success (Buisson et al. 2020).

Applying the Framework for Restoration of the Invaded *Shola*-Grassland Ecosystem in South India

The *shola*-grassland ecosystem, a mosaic of montane cloud forests (locally called *sholas*) in valleys and sheltered areas, and C₄ grasslands on exposed slopes and hilltops, occurs in montane regions of the Western Ghats biodiversity hotspot in southern India. In the Nilgiri Biosphere Reserve (Bunyan et al. 2012), these mosaics occur at elevations ranging from 1,700–2,400 m a.s.l (above sea level), and receive average annual rainfall amounts of 2,500–5,600 mm (Caner et al. 2007). Paleo-ecological studies indicate that this mosaic is at least 40,000 years old, albeit dynamic, with grasslands expanding during colder and drier periods (Rajagopalan et al. 1997). Frost is an important driver maintaining this mosaic currently by killing *shola* tree seedlings in grasslands (Joshi et al. 2020). These grasslands provide habitats for many endangered and endemic species (Thomas & Palmer 2007), and play a crucial role in water regulation (Sikka et al. 2003).

At least 40 exotic plant species were introduced in the Nilgiris in the 19th and early 20th centuries, of which three woody N-fixers have become invasive in the montane grasslands

(*Acacia mearnsii*, *Cytisus scoparius*, and *Ulex europaeus*; Joshi et al. 2018; Sriramamurthy et al. 2020). Invasion by these woody N-fixers has decreased the area of the Nilgiri grasslands by 50–60% from 1973 to 2017 (Arasumani et al. 2019), and experimental studies predict further invasion due to climate change (Joshi et al. 2020). These species decrease native diversity (Srinivasan et al. 2007), reduce water yield from catchments (Sikka et al. 2003), and increase soil total N content, inorganic N availability, and mineralization rates (Raghurama 2013; Raghurama & Sankaran, unpublished data).

On the directions of a recent state high-court public interest litigation, the state forest departments are manually clearing invasives (Arasumani et al. 2021). However, these invasives produce large seed banks, allowing them to rapidly recolonize many cleared areas (Unkule 2017, personal observation). In other cleared areas, unpalatable native and exotic grasses dominate (Bosco, personal communication, 6 December 2020). Below, we apply our framework to the problem of restoring the invaded Nilgiri grasslands, and identify urgent research questions whose answers will help guide restoration efforts (Fig. 5).

Restoration targets should consider species co-occurrences in different environments in the Nilgiri grasslands (Srinivasan et al. 2015), and should aim to restore not only diverse grasslands, but also ones that provide other key services such as water regulation, and are resilient to fire and herbivory (I.1.). These endogenous disturbances have been a part of these grasslands for at least the past 1,000 years, with the indigenous people of the region (Todas) using fires to stimulate fresh grass growth for their cattle until recently (Srinivasan 2012).

Since eradication of woody invasives like Acacias in grasslands is expensive due to their wide-ranging impacts and large and long-lived seed banks (Le Maitre et al. 2011), areas with low-invasion intensity should be prioritized for restoration in the Nilgiris (Arasumani et al. 2021). Using remote sensing to incorporate other factors such as feasibility (e.g. site accessibility) and habitat connectivity in identifying priority areas, and identifying areas where passive restoration can be effective (e.g. areas with low-invasion intensity, with remaining native USOs, and with nearby native populations) can help improve the cost-effectiveness of restoration (I.2., I.3.).

Fire, in combination with post-fire management to prevent re-invasion (Le Maitre et al. 2011), can potentially be a useful tool to cost-effectively clear and manage invasion by *A. mearnsii* and *C. scoparius* in the Nilgiris, which demonstrate high adult mortality post-fire (I.4., Srinivasan et al. 2012; Sriramamurthy et al. 2020). Further, inorganic N-availability and N-mineralization rates in burnt areas drop to levels comparable to uninvaded grasslands (M. Raghurama & M. Sankaran, unpublished data). However, USOs of *U. europaeus* regenerate post-fire with coppicing individuals flowering within a year following fire in the Nilgiris, and fire could facilitate further invasion by this species (Sriramamurthy et al. 2020). So, fire is likely to be an effective management tool only in areas without *U. europaeus* and should be used carefully in the landscape. Where fire cannot be used, using cleared invasive biomass as fuelwood in industries can help increase the economic benefits associated with these programs.

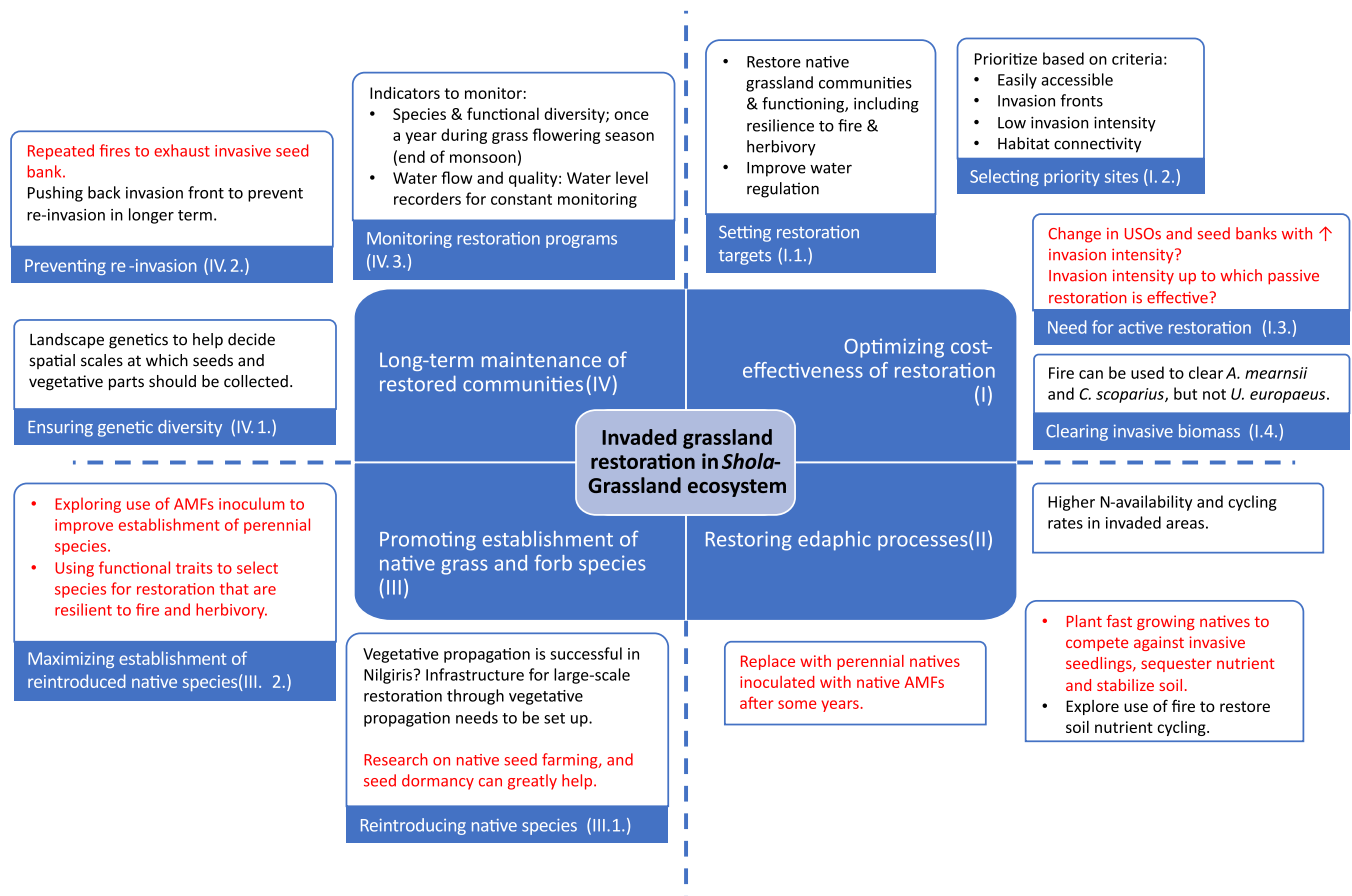


Figure 5. Applying the framework to the case of restoring the Nilgiri grasslands, invaded by three N-fixing woody species. The framework helps organize known information from previous research, and identify research priorities (in red) which can help guide the restoration of these grasslands.

The consequences of clearing these invasive plants on soil properties is however not fully understood at present. Both *A. mearnsii* and *C. scoparius* show significant legacy effects on soil properties following clearance (Unkule 2017; Slesak et al. 2021), while the impacts of *U. europaeus* appear to be potentially reversible (Bateman & Vitousek 2018). In addition to using fire to restore soil nutrient cycling where appropriate, there is also a need to explore the effectiveness of other strategies to reduce nutrient levels such as planting fast-growing native grassland species (post-clearing of invasives) to sequester nutrients in biomass, and their subsequent removal and replacement by perennial grassland species (II). Exploring the effectiveness of microbial and AMF inoculum to restore soil communities and improve restoration outcomes is also needed (II).

Presently, restoration practitioners in the Nilgiris are reintroducing native grasses largely through vegetative propagation (V. Bosco, personal communication, sixth Dec 2020) - a resource-intensive endeavor. Exploring the effectiveness of large-scale seed farming, and research into seed dormancy of native species and ways to break it, has the potential to significantly decrease restoration costs in the Nilgiri grasslands (III.1.). Detailed characterization of functional traits of native species can help ensure selection of appropriate species for

restoring relevant functions, and to identify species resilient to fire and herbivory (III.2.). Understanding landscape genetics of native grass and forb populations will help decide the spatial scales over which seeds and vegetative parts should be collected to ensure high genetic diversity in restored communities (IV.1.). Finally, evaluating impact of repeated fires post-clearance on invasive seed bank (IV.2.) is needed to assess whether prescribed burning can be used to push back the invasion front (seed sources for re-invasion) and ensure that restored areas are maintained and not re-invaded in the longer term (I.2., IV.2.). Monitoring species and functional diversity in restoration sites should occur once a year during the grass flowering season (end of the monsoon season), while installing water level recorders can help measure changes in water flow quality and quality post-restoration (IV.3.).

In summary, based on our framework, we recommend that both manual clearing and fire can be used to clear *A. mearnsii* and *C. scoparius*, but not *U. europaeus*, for which manual clearing seems to be the only feasible route (Fig. 5). Planting of native grassland species is required in areas where passive restoration is not possible. Research to determine the suite of native species and traits to be re-introduced to minimize re-invasion is required (Fig. 5). Further, post-fire de-weeding of invasive

seedlings is required in the case of usage of fire to clear *A. mearnsii* and *C. scoparius*.

Conclusion

We present a framework to guide the planning and execution of restoration of grasslands invaded by woody exotics in tropical FG mosaics. We apply this framework to the montane Nilgiri grasslands and identify urgent research efforts that can help guide the restoration of these grasslands. The framework can also help identify potential activities to significantly increase the cost-effectiveness of these programs, such as (1) choosing priority sites based on invasion intensity and factors determining feasibility (e.g. site accessibility), (2) identifying areas where passive restoration can be effective, (3) identifying opportunities for using fire to clear invasives and manage them, and (4) exploring the use of large-scale seed farming to re-introduce native species. Finally, we urge that to facilitate long-term maintenance of restored grassland communities in tropical FG mosaics, it is important to understand the environmental drivers of the mosaic, and target grassland restoration in topographical areas where grasslands naturally occur. To truly realize the potential of the UN Decade on Ecosystem Restoration, ecosystem-specific frameworks are needed to guide restoration, and we hope this framework will stimulate studies to better inform restoration of invaded tropical FG mosaics.

Acknowledgments

We thank A. Das, A. Joshi, C. Sheth, C. Dayani, D. Kaikho, K. Teegalapalli, M. Kohli, R. Rastogi, R. Sriramamurthy, S. Pulla, and S. Ongole for helpful comments on various stages of the manuscript preparation. We acknowledge the core support provided to NCBS, TIFR by the Department of Atomic Energy, Government of India, under Project Identification No. RTI 4006.

LITERATURE CITED

- Arasumani M, Khan D, Vishnudas CK, Muthukumar M, Bunyan M, Robin VV (2019) Invasion compounds an ecosystem-wide loss to afforestation in the tropical grasslands of the Shola Sky Islands. *Biological Conservation* 230:141–150
- Arasumani M, Bunyan M, Robin VV (2021) Opportunities and challenges in using remote sensing for invasive tree species management, and in the identification of restoration sites in tropical montane grasslands. *Journal of Environmental Management* 280:111759
- Bader MY, Ruijten JJ (2008) A topography-based model of forest cover at the alpine tree line in the tropical Andes. *Journal of Biogeography* 35:711–723
- Bateman JB, Vitousek PM (2018) Soil fertility response to *Ulex europaeus* invasion and restoration efforts. *Biological Invasions* 20:2777–2791
- Biondo R, Souza AV, Bertoni BW, Soares AM, França SC, Pereira AMS (2007) Micropropagation, seed propagation and germplasm bank of *Mandevilla velutina* (Mart.) Woodson. *Scientia Agricola* 64:263–268
- Blair JM (1997) Fire, N availability, and plant response in grasslands: a test of the transient maxima hypothesis. *Ecology* 78:2359–2368
- Bond WJ (2019) Open ecosystems: ecology and evolution beyond the forest edge. Oxford, United Kingdom: Oxford University Press
- Bond WJ, Parr CL (2010) Beyond the forest edge: ecology, diversity and conservation of the grassy biomes. *Biological Conservation* 143:2395–2404
- Bond WJ, Silander JA Jr, Ranaivonasy J, Ratsirason J (2008) The antiquity of Madagascar's grasslands and the rise of C4 grassy biomes. *Journal of Biogeography* 35:1743–1758
- Brooks ML, D'antonio CM, Richardson DM, Grace JB, Keeley JE, DiTomaso JM, Hobbs RJ, Pellant M, Pyke D (2004) Effects of invasive alien plants on fire regimes. *Bioscience* 54:677–688
- Bucharova A, Michalski S, Hermann JM, Heveling K, Durka W, Hölzel N, Kollmann J, Bossdorf O (2017) Genetic differentiation and regional adaptation among seed origins used for grassland restoration: lessons from a multispecies transplant experiment. *Journal of Applied Ecology* 54:127–136
- Buisson E, Le Stradic S, Silveira FA, Durigan G, Overbeck GE, Fidelis A, et al. (2019) Resilience and restoration of tropical and subtropical grasslands, savannas, and grassy woodlands. *Biological Reviews* 94:590–609
- Buisson E, Fidelis A, Overbeck GE, Schmidt IB, Durigan G, Young TP, et al. (2020) A research agenda for the restoration of tropical and subtropical grasslands and savannas. *Restoration Ecology* 29:e13292. <https://doi.org/10.1111/rec.13292>
- Bunyan M, Bardhan S, Jose S (2012) The shola (tropical montane forest)-grassland ecosystem mosaic of peninsular India: a review. *American Journal of Plant Sciences* 3:1632–1639
- Caner L, Seen DL, Gunnell Y, Ramesh BR, Bourgeon G (2007) Spatial heterogeneity of land cover response to climatic change in the Nilgiri highlands (southern India) since the Last Glacial Maximum. *The Holocene* 17:195–205
- Cleland EE, Larios L, Suding KN (2013) Strengthening invasion filters to reassemble native plant communities: soil resources and phenological overlap. *Restoration Ecology* 21:390–398
- Coutinho AG, Alves M, Sampaio AB, Schmidt IB, Vieira DLM (2019) Effects of initial functional-group composition on assembly trajectory in savanna restoration. *Applied Vegetation Science* 22:61–70
- Dayrell RL, Arruda AJ, Buisson E, Silveira FA (2016) Overcoming challenges on using native seeds for restoration of megadiverse resource-poor environments: a reply to Madsen et al. *Restoration Ecology* 24:710–713
- DiTomaso JM, Brooks ML, Allen EB, Minnich R, Rice PM, Kyser GB (2006) Control of invasive weeds with prescribed burning. *Weed Technology* 20:535–548
- Dudley N, Eufemia L, Fleckenstein M, Periago ME, Petersen I, Timmers JF (2020) Grasslands and savannas in the UN Decade on Ecosystem Restoration. *Restoration Ecology* 28:1313–1317
- Durbecq A, Jaunatre R, Buisson E, Cluchier A, Bischoff A (2020) Identifying reference communities in ecological restoration: the use of environmental conditions driving vegetation composition. *Restoration Ecology* 28:1445–1453
- Evimer VT, Hawkes CV (2008) Embracing variability in the application of plant–soil interactions to the restoration of communities and ecosystems. *Restoration Ecology* 16:713–729
- Fensham RJ, Butler DW, Fairfax RJ, Quintin AR, Dwyer JM (2016) Passive restoration of subtropical grassland after abandonment of cultivation. *Journal of Applied Ecology* 53:274–283
- Figueiredo MA, Diniz AP, Messias MCTB, Kozovits AR (2018) Propagation and establishment of rupestrian grassland grasses for restoration of degraded areas by mining. *Brazilian Journal of Botany* 41:287–295
- Funk JL, Cleland EE, Suding KN, Zavaleta ES (2008) Restoration through reassembly: plant traits and invasion resistance. *Trends in Ecology & Evolution* 23:695–703
- Fynn RW, Wragg PD, Morris CD, Kirkman KP, Naiken J (2009) Vegetative traits predict grass species' invasiveness and the invasibility of restored grassland. *African Journal of Range & Forage Science* 26:59–68
- Gann GD, McDonald T, Walder B, Aronson J, Nelson CR, Jonson J, et al. (2019) International principles and standards for the practice of ecological restoration. *Restoration Ecology* 27:S1–S46
- Gómez-Aparicio L (2009) The role of plant interactions in the restoration of degraded ecosystems: a meta-analysis across life-forms and ecosystems. *Journal of Ecology* 97:1202–1214

- Grace JB, Smith MD, Grace SL, Collins SL, Stohlgren TJ (2001) Interactions between fire and invasive plants in temperate grasslands of North America. Pages 40–65 In: Galley KEM, Wilson TP (eds) Proceedings of the invasive species workshop: the role of fire in the control and spread of invasive species. Fire conference 2000. Miscellaneous Publication No. 11, Tall Timbers Research Station, Tallahassee, FL.
- Grove S, Parker IM, Haubensak KA (2015) Persistence of a soil legacy following removal of a nitrogen-fixing invader. *Biological Invasions* 17:2621–2631
- Grove S, Haubensak KA, Gehring C, Parker IM (2017) Mycorrhizae, invasions, and the temporal dynamics of mutualism disruption. *Journal of Ecology* 105:1496–1508
- Hobbs RJ, Higgs E, Harris JA (2009) Novel ecosystems: implications for conservation and restoration. *Trends in Ecology & Evolution* 24:599–605
- Hoeksema JD, Chaudhary VB, Gehring CA, Johnson NC, Karst J, Koide RT, et al. (2010) A meta-analysis of context-dependency in plant response to inoculation with mycorrhizal fungi. *Ecology Letters* 13:394–407
- Hoffman AM, Smith MD (2018) Thinking inside the box: tissue culture for plant propagation in a key ecological species, *Andropogon gerardii*. *American Journal of Plant Sciences* 9:1987–2003
- Holmes PM, Esler KJ, Gaertner M, Geerts S, Hall SA, Nsikani MM, Richardson DM, Ruwanga S (2020) Biological invasions and ecological restoration in South Africa. Pages 665–700. In: van Wilgen BW, Measey J, Richardson DM, Wilson JR, Zengeya TA (eds) Biological invasions in South Africa. Springer International Publishing, Cham, Switzerland
- Inderjit, Van der Putten WH (2010) Impacts of soil microbial communities on exotic plant invasions. *Trends in Ecology & Evolution* 25:512–519
- Jo I, Fridley JD, Frank DA (2017) Invasive plants accelerate nitrogen cycling: evidence from experimental woody monocultures. *Journal of Ecology* 105:1105–1110
- Joshi AA, Sankaran M, Ratnam J (2018) ‘Forestry’ the grassland: historical management legacies in forest-grassland mosaics in southern India, and lessons for the conservation of tropical grassy biomes. *Biological Conservation* 224:144–152
- Joshi AA, Ratnam J, Sankaran M (2020) Frost maintains forests and grasslands as alternate states in a montane tropical forest–grassland mosaic; but alien tree invasion and warming can disrupt this balance. *Journal of Ecology* 108:122–132
- Kannan R, Shackleton CM, Krishnan S, Shaanker RU (2016) Can local use assist in controlling invasive alien species in tropical forests? The case of *Lantana camara* in southern India. *Forest Ecology and Management* 376:166–173
- Kimuti DW, Riginos C, Belnap J (2017) Low-cost grass restoration using erosion barriers in a degraded African rangeland. *Restoration Ecology* 25:376–384
- Kozioł L, Bever JD (2015) Mycorrhizal response trades off with plant growth rate and increases with plant successional status. *Ecology* 96:1768–1774
- Kozioł L, Bever JD (2017) The missing link in grassland restoration: arbuscular mycorrhizal fungi inoculation increases plant diversity and accelerates succession. *Journal of Applied Ecology* 54:1301–1309
- Kozioł L, Schultz PA, House GL, Bauer JT, Middleton EL, Bever JD (2018) The plant microbiome and native plant restoration: the example of native mycorrhizal fungi. *Bioscience* 68:996–1006
- Kull CA, Shackleton CM, Cunningham PJ, Ducatillon C, Dufour-Dror JM, Esler KJ, et al. (2011) Adoption, use and perception of Australian acacias around the world. *Diversity and Distributions* 17:822–836
- Le Maitre DC, Gaertner M, Marchante E, Ens EJ, Holmes PM, Pauchard A, O’Farrell PJ, Rogers AM, Blanchard R, Blignaut Richardson DM (2011) Impacts of invasive Australian acacias: implications for management and restoration. *Diversity and Distributions* 17:1015–1029
- León Cordero R (2016) The invasion of forest-grassland mosaics by gorse, *Ulex europaeus* L., Fabaceae: multi-scale analyses of landscape, community structure and anthropogenic effects. PhD thesis, University of Guelph, Guelph, Ontario, Canada
- Madsen MD, Davies KW, Boyd CS, Kerby JD, Svejcar TJ (2016) Emerging seed enhancement technologies for overcoming barriers to restoration. *Restoration Ecology* 24:S77–S84
- Manzano S, Julier AC, Dirk CJ, Razafimanantsoa AH, Samuels I, Petersen H, Gell P, Hoffman MT, Gillson L (2020) Using the past to manage the future: the role of palaeoecological and long-term data in ecological restoration. *Restoration Ecology* 28:1335–1342
- McKay JK, Christian CE, Harrison S, Rice KJ (2005) “How local is local?”—a review of practical and conceptual issues in the genetics of restoration. *Restoration Ecology* 13:432–440
- Midgley JJ, Lawes MJ, Chamaillé-Jammes S (2010) Savanna woody plant dynamics: the role of fire and herbivory, separately and synergistically. *Australian Journal of Botany* 58:1–11
- Nangendo G, Stein A, Gelens M, de Gier A, Albricht R (2002) Quantifying differences in biodiversity between a tropical forest area and a grassland area subject to traditional burning. *Forest Ecology and Management* 164:109–120. [https://doi.org/10.1016/s0378-1127\(01\)00603-x](https://doi.org/10.1016/s0378-1127(01)00603-x)
- Nerlekar AN, Veldman JW (2020) High plant diversity and slow assembly of old-growth grasslands. *Proceedings of the National Academy of Sciences* 117:18550–18556
- Nsikani MM, van Wilgen BW, Gaertner M (2018) Barriers to ecosystem restoration presented by soil legacy effects of invasive alien N₂-fixing woody species: implications for ecological restoration. *Restoration Ecology* 26:235–244
- Nsikani MM, Gaertner M, Kritzing-Klopper S, Ngubane NP, Esler KJ (2019) Secondary invasion after clearing invasive *Acacia saligna* in the South African fynbos. *South African Journal of Botany* 125:280–289
- de Oliveira AC, Forti VA, de Paula Lioila P, Viani RA (2020) Techniques for seedling production of two native grasses: new perspectives for Brazilian Cerrado restoration. *Restoration Ecology* 28:297–303
- Overbeck GE, Müller SC, Fidelis A, Pfadenhauer J, Pillar VD, Blanco CC, Boldrini II, Both R, Forneck ED (2007) Brazil’s neglected biome: the south Brazilian Campos. *Perspectives in Plant Ecology, Evolution and Systematics* 9:101–116
- Parr CL, Lehmann CE, Bond WJ, Hoffmann WA, Andersen AN (2014) Tropical grassy biomes: misunderstood, neglected, and under threat. *Trends in Ecology & Evolution* 29:205–213
- Pearson DE, Ortega YK, Runyon JB, Butler JL (2016) Secondary invasion: the bane of weed management. *Biological Conservation* 197:8–17
- Perkins LB, Hatfield G (2016) Can commercial soil microbial treatments remediate plant–soil feedbacks to improve restoration seedling performance? *Restoration Ecology* 24:194–201
- Perring MP, Standish RJ, Price JN, Craig MD, Erickson TE, Ruthrof KX, Whiteley AS, Valentine LE, Hobbs RJ (2015) Advances in restoration ecology: rising to the challenges of the coming decades. *Ecosphere* 6:1–25
- Perry LG, Blumenthal DM, Monaco TA, Paschke MW, Redente EF (2010) Immobilizing nitrogen to control plant invasion. *Oecologia* 163:13–24
- Pokorny ML, Sheley RL, Zabinski CA, Engel RE, Svejcar TJ, Borkowski JJ (2005) Plant functional group diversity as a mechanism for invasion resistance. *Restoration Ecology* 13:448–459
- Prach K, Šebelíková L, Řehounková K, del Moral R (2019a) Possibilities and limitations of passive restoration of heavily disturbed sites. *Landscape Research* 45:247–253
- Prach K, Durigan G, Fennessy S, Overbeck GE, Torezan JM, Murphy SD (2019b) A primer on choosing goals and indicators to evaluate ecological restoration success. *Restoration Ecology* 27:917–923
- Prober SM, Byrne M, McLean EH, Steane DA, Potts BM, Vaillancourt RE, Stock WD (2015) Climate-adjusted provenancing: a strategy for climate-resilient ecological restoration. *Frontiers in Ecology and Evolution* 3:65
- Raghurama M (2013) Variation in the sequestration of carbon and nitrogen in soils with change in vegetation and depth: a study in the Nilgiri Mountains, India. MSc thesis, Birla Institute of Technology and Science, Pilani, Rajasthan, India
- Rajagopalan G, Sukumar R, Ramesh R, Pant RK, Rajagopalan G (1997) Late quaternary vegetational and climatic changes from tropical peats in southern India – an extended record up to 40,000 years BP. *Current Science* 73:60–63
- Rappaport DI, Tambosi LR, Metzger JP (2015) A landscape triage approach: combining spatial and temporal dynamics to prioritize restoration and conservation. *Journal of Applied Ecology* 52:590–601

- Rogers D, McGuire P (2015) Genetic erosion: context is key. Pages 1–24. In: Ahuja MR, Jain SM (eds) Genetic diversity and erosion in plants. Springer, Cham
- Sardans J, Bartrons M, Margalef O, Gargallo-Garriga A, Janssens IA, Ciais P, Obersteiner M, Sigurdsson BD, Chen HYH, Peñuelas J (2017) Plant invasion is associated with higher plant–soil nutrient concentrations in nutrient-poor environments. *Global Change Biology* 23:1282–1291
- Sikka AK, Samra JS, Sharda VN, Samraj P, Lakshmanan V (2003) Low flow and high flow responses to converting natural grassland into bluegum (*Eucalyptus globulus*) in Nilgiris watersheds of South India. *Journal of Hydrology* 270:12–26
- Silveira FA, Arruda AJ, Bond W, Durigan G, Fidelis A, Kirkman K, et al. (2020) Myth-busting tropical grassy biome restoration. *Restoration Ecology* 28:1067–1073
- Slesak RA, Harrington TB, D'Amato AW, Carter DR (2021) Legacy effects of non-native *Cytisus scoparius* in glacial outwash soils: potential impacts to forest soil productivity in western Washington. *Forest Ecology and Management* 481:118733
- Sommerville K, Rossetto M, Pulkownik A (2013) Maximising adaptive potential in translocated populations of clonal saltmarsh plants: a case study on *Wilsonia backhousei*, Convolvulaceae. *Wetlands Ecology and Management* 21:339–351
- Srinivasan MP (2012) Exotic shrub invasion in a montane grassland: the role of fire as a potential restoration tool. *Biological Invasions* 14:1009–1028
- Srinivasan MP, Shenoy K, Gleeson SK (2007) Population structure of scotch broom (*Cytisus scoparius*) and its invasion impacts on the resident plant community in the grasslands of Nilgiris, India. *Current Science* 93:1108–1113
- Srinivasan MP, Kalita R, Gurung IK, Bhattacharjee SK, Antony PM, Krishnan S, Gleeson SK (2012) Seedling germination success and survival of the invasive shrub scotch broom (*Cytisus scoparius*) in response to fire and experimental clipping in the montane grasslands of the Nilgiris, South India. *Acta Oecologica* 38:41–48
- Srinivasan MP, Bhatia S, Shenoy K (2015) Vegetation–environment relationships in a south Asian tropical montane grassland ecosystem: restoration implications. *Tropical Ecology* 56:201–217
- Sriramamurthy RT, Bhalla RS, Sankaran M (2020) Fire differentially affects mortality and seedling regeneration of three woody invaders in forest–grassland mosaics of the southern Western Ghats, India. *Biological Invasions* 22:1623–1634
- Stanturf JA, Palik BJ, Dumroese RK (2014) Contemporary forest restoration: a review emphasizing function. *Forest Ecology and Management* 331:292–323
- Strassburg BB, Beyer HL, Crouzeilles R, Iribarrem A, Barros F, de Siqueira MF, et al. (2019) Strategic approaches to restoring ecosystems can triple conservation gains and halve costs. *Nature Ecology & Evolution* 3:62–70
- Tambosi LR, Martensen AC, Ribeiro MC, Metzger JP (2014) A framework to optimize biodiversity restoration efforts based on habitat amount and landscape connectivity. *Restoration Ecology* 22:169–177
- Temperton VM, Buchmann N, Buisson E, Durigan G, Kazmierczak Ł, Perring MP, Dechoum MS, Veldman JW, Overbeck GE (2019) Step back from the forest and step up to the Bonn challenge: how a broad ecological perspective can promote successful landscape restoration. *Restoration Ecology* 27:705–719
- Thomas SM, Palmer MW (2007) The montane grasslands of the Western Ghats, India: community ecology and conservation. *Community Ecology* 8:67–73
- du Toit JC (2009) Early survival and growth of vegetatively propagated indigenous grasses in a clear-felled timber plantation in KwaZulu-Natal, South Africa. *African Journal of Range & Forage Science* 26:97–101
- Tsujii PK, Coutinho AG, Medina AM, Kraft NJ, Gonzalez-Melo A, Higuchi P, et al. (2020) How many and which species to plant? A multi-trait-based approach to select species to restore ecosystem services. *BioRxiv*. <https://doi.org/10.1101/2020.10.29.359455>
- Unkule M (2017) Effect of removal of an invasive species, *Acacia mearnsii* on the plant diversity in a shola-grassland ecosystem. MSc thesis, Indian Institute of Science Education and Research, Pune, India
- Veldman JW, Buisson E, Durigan G, Fernandes GW, Le Stradic S, Mahy G, et al. (2015a) Toward an old-growth concept for grasslands, savannas, and woodlands. *Frontiers in Ecology and the Environment* 13:154–162
- Veldman JW, Overbeck GE, Negreiros D, Mahy G, Le Stradic S, Fernandes GW, Durigan G, Buisson E, Putz FE, Bond WJ (2015b) Where tree planting and forest expansion are bad for biodiversity and ecosystem services. *Bioscience* 65:1011–1018
- Veldman JW, Silveira FA, Fleischman FD, Ascarrunz NL, Durigan G (2017) Grassy biomes: an inconvenient reality for large-scale forest restoration? A comment on the essay by Chazdon and Laestadius. *American Journal of Botany* 104:649–651
- Vilà M, Espinar JL, Hejda M, Hulme PE, Jarošík V, Maron JL, Pergl J, Schaffner U, Sun Y, Pyšek P (2011) Ecological impacts of invasive alien plants: a meta-analysis of their effects on species, communities and ecosystems. *Ecology Letters* 14:702–708
- Villa F, Bagstad KJ, Voigt B, Johnson GW, Portela R, Honzák M, Batker D (2014) A methodology for adaptable and robust ecosystem services assessment. *PLoS One* 9:e91001
- Vitousek PM, Walker LR (1989) Biological invasion by *Myrica faya* in Hawai'i: plant demography, nitrogen fixation, ecosystem effects. *Ecological Monographs* 59:247–265
- Wagenius S, Beck J, Kiefer G (2020) Fire synchronizes flowering and boosts reproduction in a widespread but declining prairie species. *Proceedings of the National Academy of Sciences* 117:3000–3005
- Wakeling JL, Cramer MD, Bond WJ (2012) The savanna-grassland 'treeline': why don't savanna trees occur in upland grasslands? *Journal of Ecology* 100:381–391
- Warman L, Moles AT (2009) Alternative stable states in Australia's Wet Tropics: a theoretical framework for the field data and a field-case for the theory. *Landscape Ecology* 24:1–13
- Wigley BJ, Charles-Dominique T, Hempson GP, Stevens N, TeBeest M, Archibald S, et al. (2021) A handbook for the standardised sampling of plant functional traits in disturbance-prone ecosystems, with a focus on open ecosystems. *Australian Journal of Botany* 68:473–531
- van Wilgen BW, Richardson DM (2014) Challenges and trade-offs in the management of invasive alien trees. *Biological Invasions* 16:721–734
- Williams SL, Davis CA (1996) Population genetic analyses of transplanted eelgrass (*Zostera marina*) beds reveal reduced genetic diversity in southern California. *Restoration Ecology* 4:163–180
- Wilson KA, Lulow M, Burger J, Fang YC, Andersen C, Olson D, O'Connell M, McBride MF (2011) Optimal restoration: accounting for space, time and uncertainty. *Journal of Applied Ecology* 48:715–725
- Woodford DJ, Richardson DM, MacIsaac HJ, Mandrak NE, Van Wilgen BW, Weyl OL (2016) Confronting the wicked problem of managing biological invasions. *NeoBiota* 31:63–86
- Wortley L, Hero JM, Howes M (2013) Evaluating ecological restoration success: a review of the literature. *Restoration Ecology* 21:537–543
- Zhao Y, Liu Z, Wu J (2020) Grassland ecosystem services: a systematic review of research advances and future directions. *Landscape Ecology* 35:793–814